

Rohit Barnwal

IIT Delhi | Backend & GenAI Engineer | ex-Zomato, Inria | 7+ Yrs in Scalable Systems & APIs

Contact

9718439128

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Education

Integrated M.Tech in Mathematics and Computing

IIT Delhi
2009 - 2014

Research Internship (3 Credits)

University of Bordeaux, France
May 2013 – August 2013

Links

LinkedIn

GitHub

StackOverflow

About Me

Technical Skills

Programming Languages: Python, PHP, Java, Node.js, JavaScript, HTML

Frameworks & Libraries: Django, Flask, Webapp2, FastAPI, Gradio

Databases & Caching: MySQL, PostgreSQL, MongoDB, Redis, Memcache, NDB

Big Data: Hadoop, Hive, Spark, Kafka, Sqoop, Presto

Cloud & DevOps: Google Cloud Platform (GCP), Docker, GitHub, Ubuntu, CI/CD

LLMs & Generative AI: ChatGPT, Claude, Gemini, Agentic AI, MCP server, Mistral, LLaMA, Sarvam LLM

APIs & Tools: REST APIs, FAISS, TogetherAI, Colab, HuggingFace

Soft Skills

Leadership, Team Management, Critical Thinking, Ethical Decision Making, Policy Analysis, Structured Communication, Self Discipline, Adaptive Learning

Master's Thesis

Fractal Usage for Financial Time Series and Multifractal Model Implementation

Supervisor: Dr. Amit Priyadarshni

Investigated the application of fractal geometry in financial time series by computing the Hausdorff dimension for complex structures such as Mandelbrot sets. Critically analyzed the Efficient Market Hypothesis and demonstrated how the Fractal Market Hypothesis offers a more robust framework for modeling market behavior and price dynamics under real-world volatility.

Professional Summary

Backend Engineer with 7+ years of experience building scalable web applications and distributed systems. Skilled in Python, Java, Node.js, and PHP with expertise in RESTful APIs, microservices, cloud platforms (AWS, GCP) and Gen AI. Proven track record of optimizing system performance, designing robust architecture, and delivering production-ready solutions in cross-functional teams.

Professional Experience

Freelance

Outlier, Upwork, Remote

April 2025 – Present

- Designed and built Digital Twin – an AI-powered journaling agent using SentenceTransformers, FAISS, and Mistral LLM to reflect personal memory and identity. Integrated with Telegram for self-tracking and future-ready for multimodal input and voice training.
- Created Agentic Workflow – a Telegram productivity bot that delivers daily prompts and logs reflection data. Built using Python, Uvicorn, and Pydantic with analytics ready backend.
- Developed a Django-based UPSC Preparation Dashboard to track syllabus coverage, resource, task lists, and document upload. Deployed at <http://loststar.pythonanywhere.com>.
- Evidence-Aware Fact Verifier- Developed an LLM-powered system to verify user claims against online sources using search tools, evidence extraction, and verdict generation with citations.
- Built an end-to-end OCR + LLM system to extract text from scanned documents, parse their structure, and generate summaries or answers. Supports resume parsing, invoice extraction, and academic paper classification.

Web Developer & System Architect

Unravel Events Internet Services Pvt Ltd, Remote

May 2019 – April 2022

- Designed and implemented AI-powered automated seller onboarding system, enabling 100,000+ sellers onboarded in 4 months using TensorFlow.
- Developed high-performance web and mobile apps achieving <300ms load times.
- Architected scalable AWS & GCP infrastructure to support real-time bookings and high concurrency.
- Reduced manual onboarding effort by 90% with intelligent face, document, and signature recognition solutions.
- Integrated innovative referral system driving rapid user and seller acquisition and sustained growth.

Senior Software Engineer

Zomato, Gurgaon

April 2017 – March 2019

- Built and maintained table reservation system, contributing to a 60% increase in restaurants accepting online reservations and 3.5x year-over-year growth in bookings.
- Scaled web applications and improved reliability using PHP, Redis, Memcache, AWS, and REST APIs.
- Spearheaded the formulation, documentation, and implementation of scalable development plans for large-scale web platforms.

Data Engineer

Zomato, Gurgaon

December 2018 – March 2019

- Designed ETL frameworks using Hive and Sqoop to move data across MySQL, HBase, and S3.
- Built Presto plugins and automated data workflows for improved data processing efficiency.

Software Developer

MyPAT- FIITJEE, New Delhi

Jan 2017 – March 2017

- Developed performance dashboards, analytics modules, test report and heatmap generation features for student tracking.

Implemented simulations using MATLAB to visualize multifractal structures and volatility clustering. Findings were presented at the IIT Delhi internal research colloquium (2014).

Research Experience

Research Intern at Cepage Team, Institut de mathematiques and Inria, France

May 2013 – August 2013

- Designed a resilient VM allocation heuristic to optimize cloud resource use under dynamic workloads.
- Achieved 66% utilization while minimizing SLA violations.

Awards

Charpak Scholarship, French Embassy

Interests

Reading, Philosophy, Watching Tennis

Software Engineer

TheSquare, New Delhi

August 2014 – January 2017

- Designed and maintained core infrastructure using Python and Google App Engine
- Integrated global payment gateways (HSBC Global Iris, PayPal), enabling smooth, secure transactions.
- Implemented advanced search functionality with multi-level filters, improving user experience and increasing site traffic by 11%.
- Migrated database and media files from SQL Server to Google Cloud Storage, enhancing data reliability and scalability.
- Developed and documented REST APIs integrated by 3rd party applications, resulting in increased traffic by 23% in 6 months.
- Built data scraping pipelines using Scrapy and BeautifulSoup to enrich platform data.
- Created a housekeeping web app to streamline operational workflows.

Interdisciplinary Learning

Civil Services Preparation (UPSC CSE – Interview Stage)

June 2022 – April 2025

- Appeared in UPSC CSE and reached interview stage — among the top 0.2% of ~1 million candidates, demonstrating analytical depth, problem-solving, and strong understanding of economic policy and international affairs.
- Developed proficiency in document analysis (Economic Survey, India Year Book, Strategy for New India @75 – NITI Aayog) and structured multi-domain learning.
- Studied ethics, Indian and Western philosophy (Advaita, Kant, Utilitarianism, Existentialism) with application to algorithmic fairness and AI alignment.
- Explored anthropological theories on culture, identity, and cognition—applicable in user modeling and LLM behavior tuning.
- Gained policy literacy in data governance, surveillance ethics, and the socio-technical landscape of digital India.
- Practiced deep document analysis (Economic Survey, Budget, India Year Book), enhancing reasoning under complexity.
- Developed systems thinking across economics, society, governance—valuable for product vision and responsible AI design.